Mandarin Tone Coach

Adaptive Pitch Feedback for L2 Mandarin Tone Learning

Milestone 1

Video Link

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Problem & Motivation

The Challenge

Mandarin has 4 tones — the same syllable means completely different things depending on pitch shape.

mā (妈) = mother · má (麻) = hemp
mǎ (马) = horse · mà (骂) = to scold

English doesn't use pitch to change word meaning, so tones are often heard and produced incorrectly.

The Gap

Existing tools fall short:

Duolingo / HelloChinese
Audio-only feedback, no pitch visualization

Research CAPT systems
Not publicly available; designed for lab use

Praat / Commercial tools
Require phonetics training to interpret output

System Architecture

Record

Browser mic
audio-recorder-streamlit

Extract F0

Parselmouth / Praat
Silence trimming

Classify

SVM (RBF kernel)
27 acoustic features

Visualize

Pitch overlay chart
Learner vs Reference

Feedback

Plain-language cues
Tone-pair specific

Training Data

Michigan State University's Tone Perfect corpus:
9,838 monosyllabic MP3s · 6 native speakers ·
4 tones · 410 syllables

Canonical contours: Chao (1968) tone letter system ·
4 reference shapes saved as .npz

Key Design Decisions & Tradeoffs

Training Corpus Selection

Chose: Tone Perfect (monosyllabic, isolated words) — 9,838 files, 6 speakers **Why:** Classifier must work on learner's single-word recordings. Isolated syllable recordings match the inference context exactly.

⚠ **Tradeoff:** *Only 9,838 training samples — larger corpora exist but use continuous speech which introduces sentence-level F0 artifacts.*

SVM vs deep learning

Chose: SVM with RBF kernel **Why:** 9,838 training samples is too small for reliable deep learning. SVM achieves 91.95% with far less data and is interpretable.

⚠ **Tradeoff:** *A CNN trained on larger data would likely outperform, but risks overfitting with this corpus size.*

Canonical vs corpus-derived reference contours

Chose: Chao (1968) canonical tone templates **Why:** Sentence-level continuous speech F0 has natural declination that corrupts tone shape. Canonical templates from Chao (1968) are theoretically grounded and match the four tone descriptions used in every Mandarin textbook.

⚠ **Tradeoff:** *Canonical templates are theoretically grounded but may not match individual native speaker variation.*

Classifier Results — SVM, Speaker-Independent Evaluation

91.95%

Test accuracy
2 unseen speakers

MLP tested: 90.52%
SVM outperformed MLP
on this dataset size.

Per-Tone F1 Score

Main confusion: T2 ↔ T3 (16% error rate)
Both involve pitch movement — T2 rises, T3 dips then rises. Adding f0_min_pos feature improved T3 recognition in live testing.

Setup

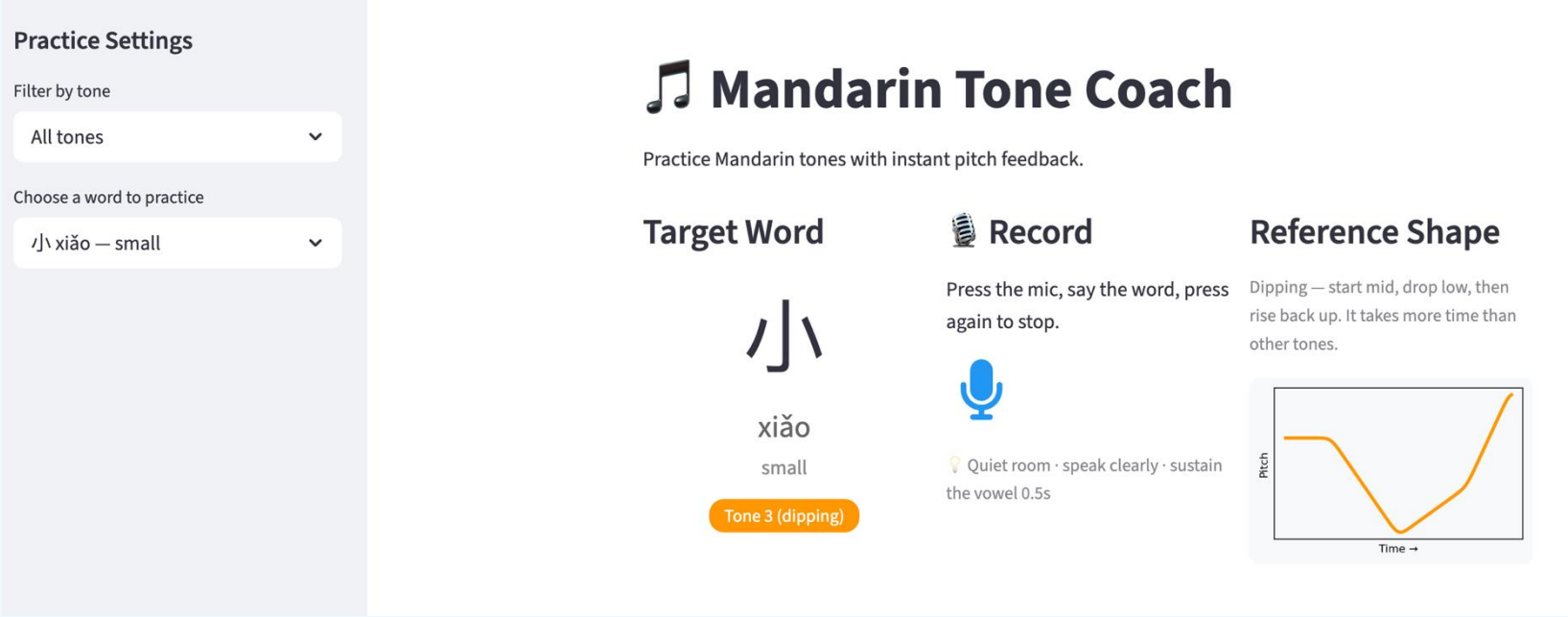
Model: SVM, RBF kernel
C=10, gamma=scale

Features: 27 per utterance
F0 contour · delta · slope
std · min/max · duration
f0_min_pos

Train: 4 speakers 6,557 samples
Test: 2 speakers 3,280 samples

Evaluation uses speaker-independent split: Female Voice 3 + Male Voice 3 held out from all training.

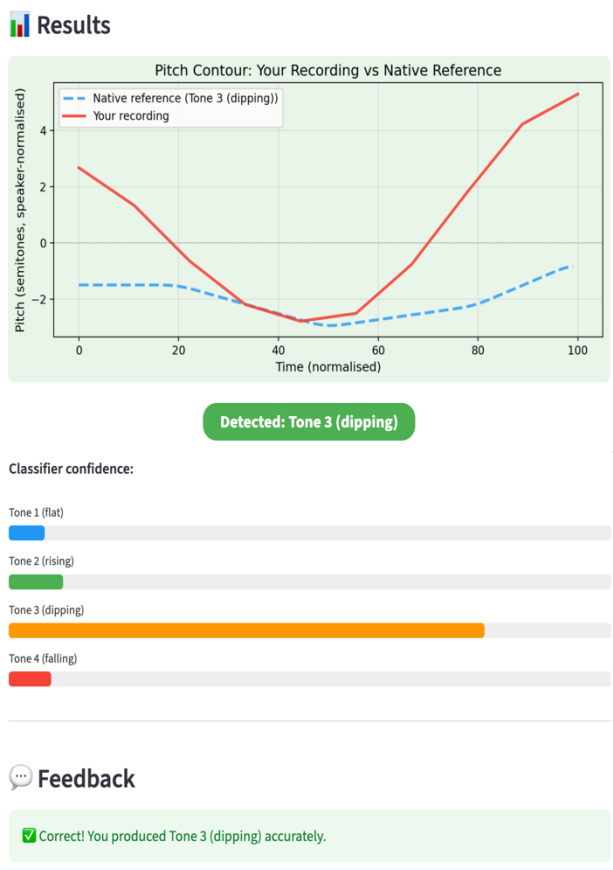
Working Prototype — Streamlit App



✅ 40 practice words across T1–T4

✅ Real-time silence trimming

Working Prototype — Streamlit App



- ✅ Pitch contour overlay chart
- ✅ Plain-language corrective feedback

Feedback Questions — For Mentor & Classmates

01

Classifier feedback design

The app currently shows the detected tone and a text explanation of what went wrong. Should it also show a numerical confidence score (e.g. '73% T3, 21% T2')? Or does that add cognitive load without pedagogical value?

02

Word list scope

The current list has 40 words (10 per tone). For beginner learners, is it more effective to practice fewer words repeatedly (e.g. 5 per tone) or to have broader vocabulary coverage?

03

T2/T3 confusion handling

16% of T2 recordings are misclassified as T3 and vice versa — a known acoustic overlap. Should the app warn the user that 'T2 and T3 are easily confused by the classifier' to calibrate expectations, or does that undermine trust in the tool?

What's Next — Weeks 11–16

Week 11 · Feedback & UI Polish

- Plain-language error cues for all T1–T4 pairs
- Disyllabic word list + tone sandhi
- UI refinements from Milestone 1 feedback

Week 12 · Evaluation Design

- Pre/post tone perception test (20 clips)
- TAM usability survey
- Consent materials & participant recruitment

Week 13 · User Study

- 6–10 participants, 1 session each
- Pre-test → practice → post-test
- Milestone 2 video submission

Wks 14–16 · Analysis & Paper

- Statistical analysis of pre/post gains
- Draft full paper
- Final submission